

Fall 2026 project assignment: Information system SmartTaxIS



SmartTaxIS is an information system for managing a taxi service focusing on modern technologies and green transportation. The system enables real-time monitoring and control of the taxi vehicles as well as order handling.

Vehicles

All vehicles managed by the system are electric and fully autonomous. The vehicles are equipped with various sensors (e.g. radars, cameras) to ensure their safe navigation. Furthermore, they are equipped with GPS and battery sensors to monitor the current position and battery charge levels. All vehicles can have additional features installed, such as entertainment systems, comfort accessories, etc. Moreover, each vehicle has a maximum passenger capacity that must not be exceeded.

A vehicle must be marked as available to be accessible for orders. The vehicle automatically sets its availability status in the system as long as it is parked in the designated place and its battery charge is above 40%.

Order placement

Registered users can order taxi transportation online. In the order, they specify the pickup and destination locations, the desired pickup time, and the number of passengers. All additional passengers must also be registered users in the system. Additionally, premium users can specify requirements for the car's equipment. Once the order is placed, the system will search for available vehicles according to the provided specification. If multiple vehicles satisfy the criteria, the following parameters are considered for selection: distance to the pickup location, the current state of the battery, and the number of completed orders in the last 3 hours. If a suitable match is found, it is offered to the user together with the expected cost and duration of the trip. If available, multiple offers can be displayed to the premium user. If there are no suitable matches, the user can repeat the search with modified parameters.

Unregistered users can do the initial search, but they cannot confirm the order until they create an account in the system. Registered users can review their order history in the system.

Passenger transportation

Once a user confirms the order, the payment is made through an external online payment provider with the user's payment card information stored in the system and the system sends the route towards the pickup location to the vehicle. Upon arrival, the system notifies the user. The vehicle will wait until all passengers get on board. If the waiting exceeds 15 minutes, the order is cancelled, and the user is charged a cancellation fee. Afterwards, the vehicle heads to the destination. A passenger can request a ride cancellation at any time.

The system will then find a suitable place to park and updates the vehicle's destination accordingly. The system maintains a list of parking spots reserved for the SmartTaxIS vehicles and monitors their availability.

Navigation

All routes to the destinations are computed by the system and then sent to the vehicle. The route is computed based on the current position of the vehicle and the traffic situation. The state of the traffic is obtained from an external traffic information provider service. The route is represented by a collection of waypoints that the autonomous car is able to follow while being able to avoid other vehicles and objects on the road on its own. The vehicle is sending its position to the system every second. If a problem occurs during the route (e.g. in case of an accident or if the vehicle cannot continue), the system operator is notified immediately. The system maintains the history of all routes, their changes, and incidents for the given vehicle for troubleshooting purposes.

Battery charging

Whenever the battery charge drops below 40%, and the vehicle is not carrying out any order at the moment, the system will navigate the vehicle to the nearest free charging station. The charging stations are managed by the system, and each station has a maximum capacity for simultaneous charging of vehicles as well as the maximum power output. Once the vehicle connects to the station, the system will enable charging. Charging is disabled once the battery charge exceeds 90%. Afterwards, the vehicle is navigated to a suitable parking spot.

Premium membership

Registered users can subscribe to a premium membership. The subscription can be billed either monthly or annually and is renewed automatically at the end of the current period. Three days before renewal, the system sends a notification to the user. The user can cancel the subscription at any time. The premium membership brings several advantages for the user, namely broader options for taxi search and a 20% discount for all orders.

Management

A system manager is able to manage (create/update/delete) all assets associated with the taxi company: employees, vehicles, charging stations, and parking spots. They can also view detailed statistics related to orders and vehicle operations.